

Adversarial Attacks on MNIST

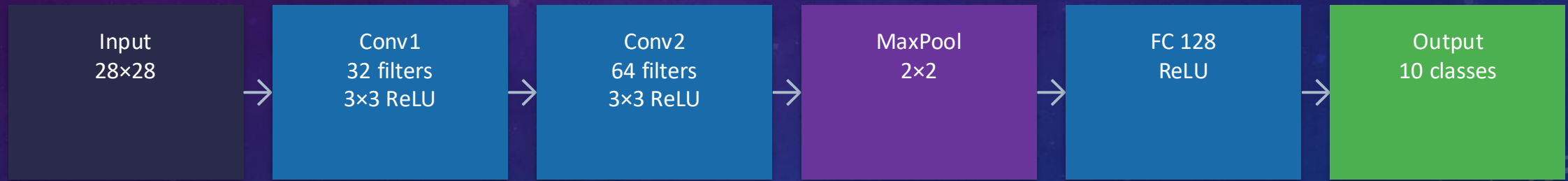
Comparing Attack Methods

Assignment 7.1 | Ziru Xu

Target Model — CNN (Trained Once, Then Attacked)

Setup

- Trained on MNIST (60k images, 5 epochs)
- Optimizer: Adam ($\text{lr} = 1\text{e-}3$) | Loss: CrossEntropyLoss
- Achieves 99.1% Recognition Rate on 10,000 clean test images
- Attacks evaluated on a 2,000-sample subset ($\epsilon = 0.3$)



Recognition Rate (RR) = 99.1% on clean test set

High clean accuracy $\neq$ adversarial robustness

Attack Methods and Metrics for Evaluation

We attack the trained CNN model using following three gradient-based adversarial attacks to evaluate the robustness of the model.

Method	Description
FGSM	Single-step gradient sign attack
I-FGSM / PGD	Multi-step iterative attack with random start
MI-FGSM	Iterative attack with momentum on accumulated gradients

We use following metrics to evaluate attack.

Metric	Definition
Recognition Rate (RR)	Accuracy on clean (unattacked) test images
Attack Success Rate (ASR)	% of correctly-classified samples that are misclassified after the attack

Three Gradient-Based Attacks ($\epsilon = 0.3$)

FGSM

Single-step gradient sign attack

- ASR: 32.6% — weakest of three
- One-step attack — fastest to compute
- $x_{adv} = x + \epsilon \cdot \text{sign}(\nabla_x L(f(x), y))$
- Single gradient call — budget not fully exploited

I-FGSM / PGD

Iterative Projected Gradient Descent

- ASR: 70.8% — strongest ★
- 50 steps ($\alpha = 0.01$), random ℓ_∞ initialisation
- Projects back to ϵ -ball after every step
- Fully exploits the perturbation budget

MI-FGSM

Momentum Iterative FGSM

- ASR: 61.5% — second strongest
- 50 steps ($\alpha = 0.01$, $\mu = 1.0$)
- Accumulates momentum: $g \leftarrow \mu g + \nabla / \|\nabla\|_1$
- Smoother gradient direction, more transferable

Results — RR & Attack Success Rate ($\epsilon = 0.3$, $n = 2,000$)

Metric	Value	Interpretation
Recognition Rate (before attacks)	99.1%	Baseline clean accuracy on 10k images
ASR — FGSM	32.6%	Weakest — single step, budget underused
ASR — MI-FGSM	61.5%	Momentum stabilises gradient direction
ASR — I-FGSM / PGD ★	70.8%	Strongest — random start + 50 projected steps <i>RR = 99.1% (clean baseline)</i>



- FGSM: 32.6% (weakest)
- MI-FGSM: 61.5% (second)
- I-FGSM/PGD: 70.8% (strongest)
- Random init > momentum here

Key Takeaways

Accuracy $\neq$ Robustness

CNN achieves 99.1% clean accuracy, yet FGSM alone fools it 32.6% of the time with perturbations invisible to humans.

Iterative attacks are far stronger than single-step

FGSM (32.6%) vs I-FGSM/PGD (70.8%). Fifty small projected steps fully exploit the $\epsilon = 0.3$ budget.

Random initialisation is the decisive factor here

PGD (70.8%) > MI-FGSM (61.5%). Random start explores a wider area of the ϵ -ball than momentum alone.

Defence is necessary

High clean accuracy cannot be trusted. Adversarial training, input purification, or certified defences are needed to close the robustness gap.